7 The distance from the Earth to the Sun is 1.496×10^{11} metres.
The speed of light is 3×10^{8} metres per second.

(a) Show that, correct to 3 significant figures, light will take 0.139 hours to travel from the Sun to the Earth.

Time = distance ÷ speed

1 mark
Distance ÷ speed

$= (1.496 \times 10^{11}) \div (3 \times 10^{8}) = 498.666666...s.$

Hours = 498.666666... ÷ 3600 = 0.1385185185...

1 mark
Seconds to hours

= 0.139 hours (3 significant figures).

1 mark
0.139 shown

(3)

1 googol is 1×10^{100}

Danesh says,

"When I multiply 1.496×10^{11} by 6.68×10^{9} I get nearly 1 googol because $1.496 \times 10^{11} \times 6.68 \times 10^{9} = 9.99 \times 10^{99}$"

(b) Is Danesh correct?
Give a reason for your answer.

No. Multiply the numbers but add the indices:

$1.496 \times 6.68 = 9.99328$; $11 + 9 = 20$.

The product is 9.99328×10^{20}, nowhere near 10^{100}.

1 mark
Add indices; power20

(1)

(Total for Question 7 is 4 marks)